

## Worksheet 1

The demand and supply for grass seed can be expressed by

$$Q_D = 100 - 2p \quad Q_S = 3p$$

where  $Q$  is the quantity of seed and  $p$  is the price. While units of measurement are valuable in analyzing markets, we will ignore those and simply treat  $p$  as the price per unit and  $Q$  as the number of units per period of time (e.g. 1000's of tons per month). Each question is worth 2 points and partial credit may be given.

1. If the market price were \$25, would there be a shortage or surplus and what would be the size of that shortage or surplus?
2. Compute the equilibrium price and quantity of Grass Seed.
3. Price elasticity is a measure of the percentage change in  $Q$  for a percentage change in price. It is calculated by the following formula:

$$E = \frac{dQ}{dp} \times \frac{p}{Q}$$

where  $\frac{dQ}{dp}$  is the slope of the demand curve (also written as  $\frac{\Delta Q}{\Delta p}$  for linear equations).

Calculate the price elasticity of demand at the equilibrium. Calculate the price elasticity of supply at the equilibrium price. Is demand *elastic*, *inelastic*, or *unit elastic*?

4. Suppose the government makes purchases of 15 units of seed. Compute the new demand curve with the increased quantity purchased by the government (Hint: add the government purchases quantity to the demand equation). (i) On a graph, draw this shift, clearly labeling the axes, the curves and the change in equilibrium from the original price and quantity to the new. Practice at drawing clear and legible graphs is important here, so make an effort to draw this neatly and clearly. (ii) Calculate the new equilibrium price and quantity. (iii) Has the equilibrium quantity increased by 15 units, more than, or less than that?
5. Instead of purchasing seed, now suppose the government imposes a specific tax on sellers of \$5 per unit sold. Compute the new equilibrium price and quantity. What is the tax revenue collected by the government? **Solved Problem 3.4** from the textbook may help you solve this if you are unsure how to proceed.
6. How much of the tax burden is born by the sellers? How much is born by the buyers?